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Dear Editorial Board,

We are pleased to submit our manuscript entitled **"Hot Wings: The Impact of Large Appendages on the Heat Loss Potential and Migratory Decision-Making of Humpback Whales"** for consideration as a Research Article in *The Journal of Experimental Biology*.

Understanding how animals regulate heat exchange across contrasting environments is a central theme in functional ecology, particularly for large-bodied endotherms operating near physiological limits. In this study, we develop a mechanistic, whole-body model of heat loss in humpback whales that explicitly incorporates the role of disproportionately large appendages as dynamic thermal windows. By combining extensive UAS-based morphometrics, biologging-derived ventilation rates, and anatomical measurements, and computational modeling, we quantify both the lower and upper bounds of heat loss across body sizes, life stages, and two dramatically different thermal environments.

Our results demonstrate that humpback whale flippers and flukes substantially expand heat dissipation capacity in warm breeding habitats while allowing effective heat conservation in cold foraging grounds. This expanded thermoregulatory scope has important consequences for sustainable metabolic range, activity levels, and early-life energetic constraints. In particular, we show that calves born in colder environments would require higher minimum metabolic rates to offset heat loss, providing new quantitative context for long-standing hypotheses about the ecological and physiological drivers of baleen whale migration.

We believe this work will be of broad interest to readers of *The Journal of Experimental Biology* because it (i) integrates morphology, physiology, and environment into a unified functional framework, (ii) advances general understanding of how body size and appendage scaling shape thermal performance, and (iii) highlights thermoregulation as a key axis linking individual physiology to large-scale ecological patterns such as migration. Beyond cetaceans, our approach and findings are relevant to comparative studies of thermal biology, energetic scope, and climate vulnerability in aquatic endotherms.

This manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the submission and declare no competing interests. Thank you for considering our manuscript. We appreciate your time and look forward to your response.

Sincerely,

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